

Introduction

We investigated ships in the area of the Kiel Canal (NOK) based on AIS (Automatic Identification System) data they send to other ships. We did a thorough analysis based on this data, and provide insights on waiting time, cargo deficit and flags of passing ships.

Method

The AIS system prevents collisions between ships. AIS-equipped ships regularly transmit navigation and voyage updates to all other ships. We analyzed historic data provided by the Danish government [2] and set up a collection of live data [3].

Results

How can ships going through the NOK be grouped?

The NOK-administration defines ship groups [1] based on length, width, and draught. Our analysis confirms these groups are effective, as they provide a concise overview of all ships.

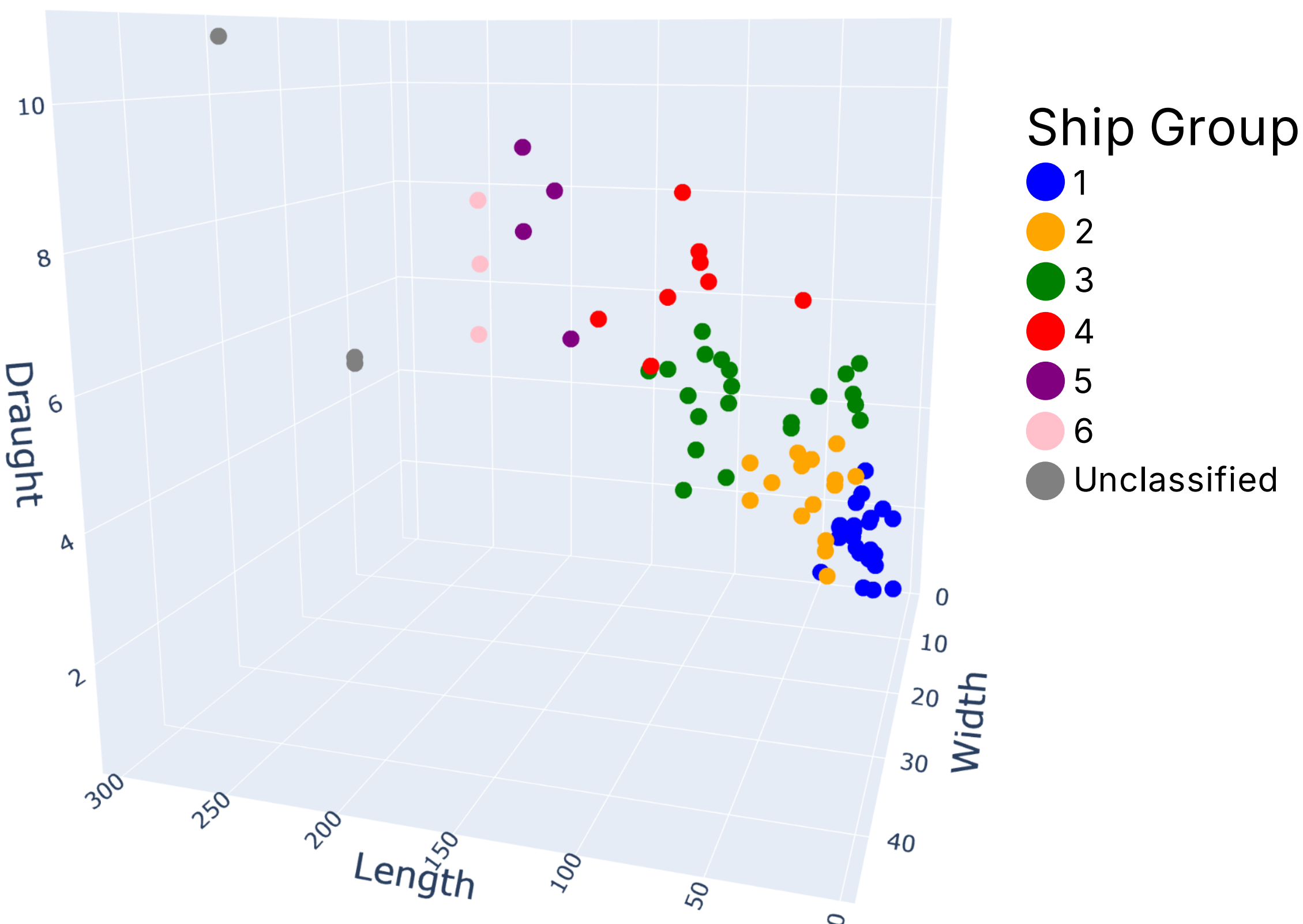


Figure 1: Passing ships grouped by the NOK administration ship groups [1]

Is there a difference between draught depending on the side of passage?

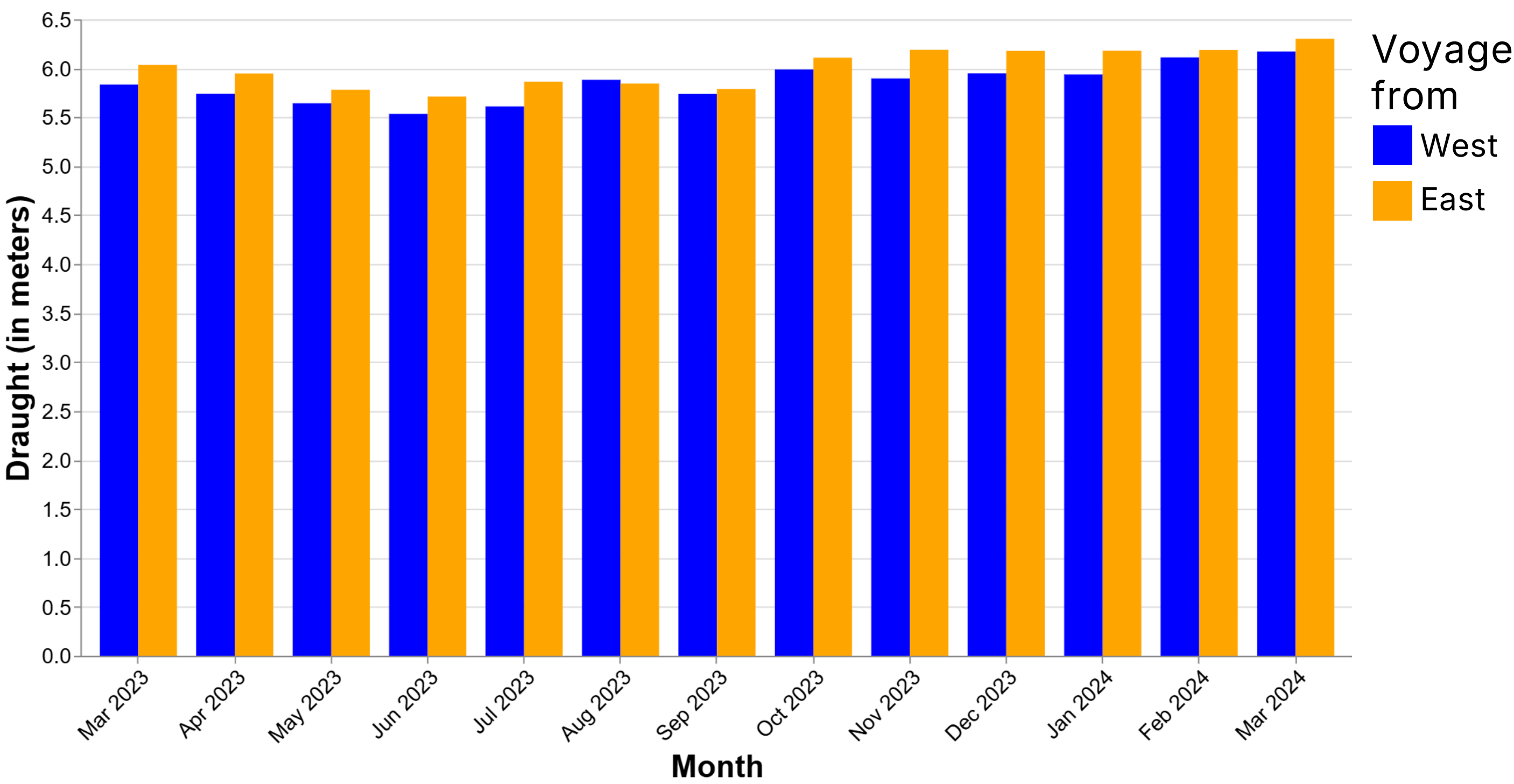


Figure 4: Mean draught difference by month and voyage direction within NOK

The higher mean draught of ships passing the Canal from the east indicates more cargo being transported from the Baltic Sea westward than vice versa.

Is there a seasonality to waiting times?

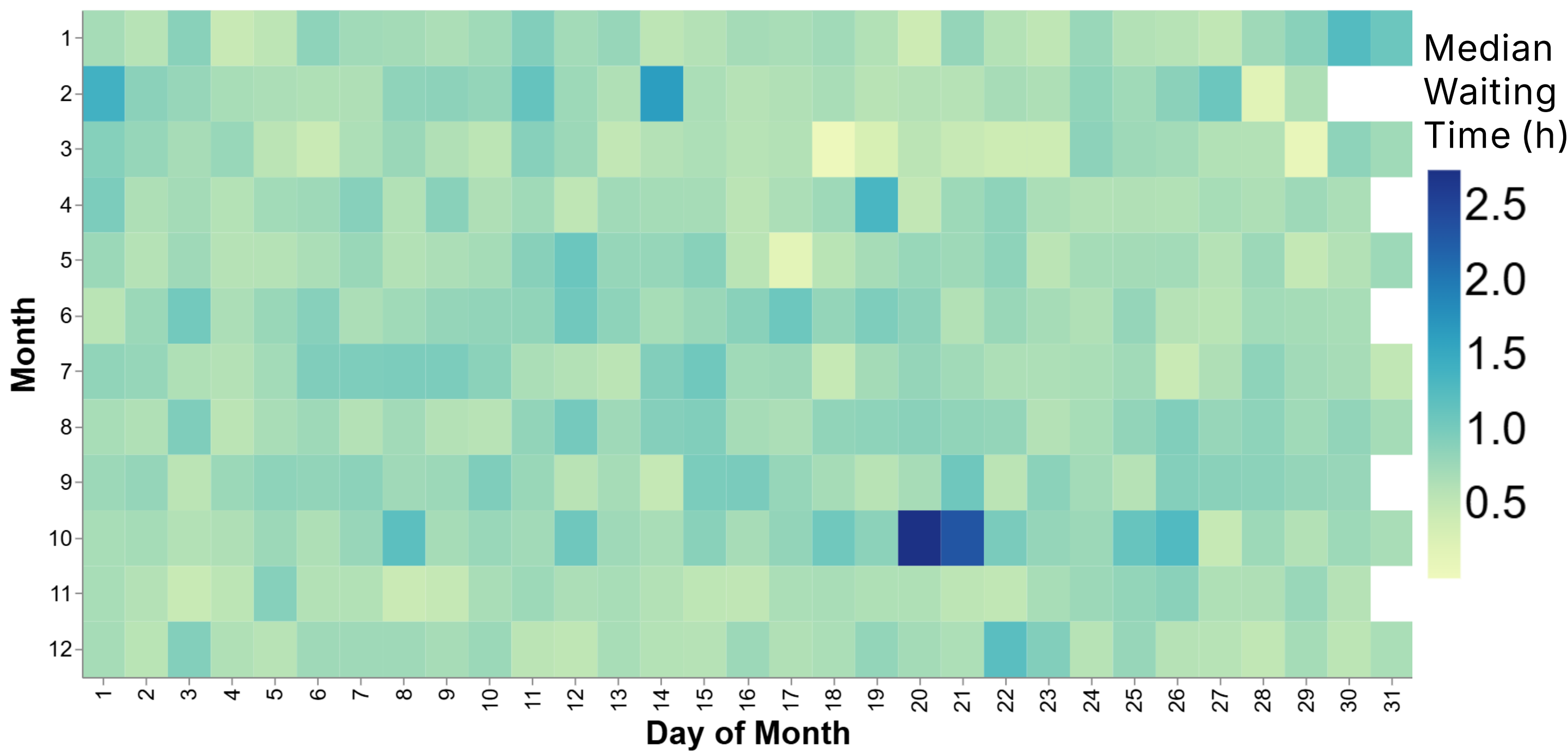


Figure 2: Median waiting time for commercial vessels grouped by date

For commercial vessels, the median waiting time decreases significantly in winter. There is a correlation between waiting time and the sailing and boating season.

How does the waiting time differ per shipgroup?

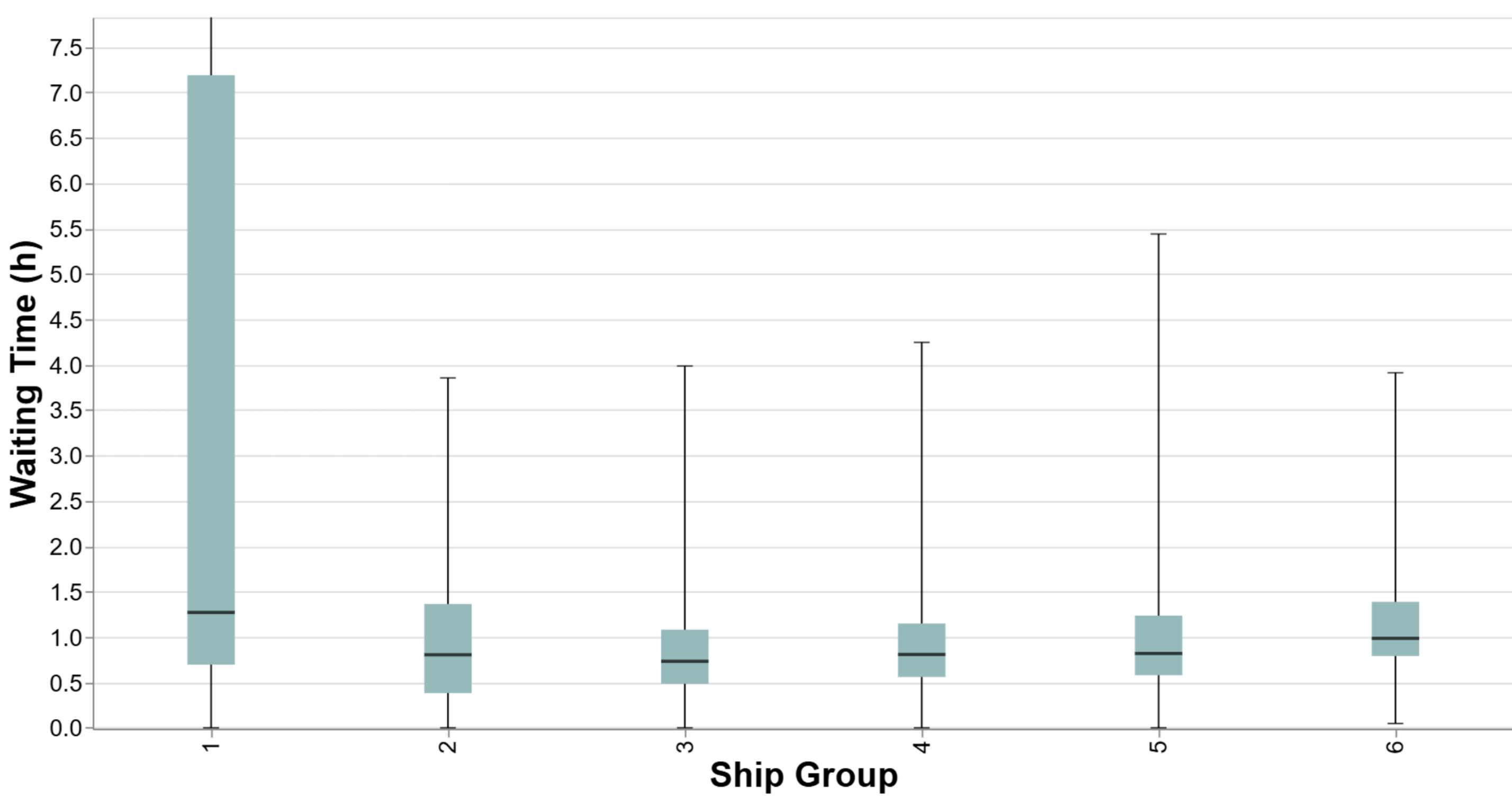


Figure 3: Whiskers plot of waiting time per ship group

The median waiting time varies only slightly per ship group. Small vessels (group 1) show the highest variance in waiting time; however the other groups also have many outliers.

What ship flags pass the NOK most commonly?



Figure 5: Map of european countries with size of points characterizing passing ships registered under their flag

Ships traversing the NOK frequently hail from neighboring countries, for instance, the Netherlands and Germany.

Nevertheless, the majority are registered under 'flags of convenience' such as Cyprus and Madeira, which are attractive due to their low taxation, streamlined registration process, and lower operational costs.

Bibliography

[1] Ship Groups defined by the federal ministry (https://www.wsa-nord-ostsee-kanal.wsv.de/Web/WSA/WSA-Nord-Ostsee-Kanal/DE/2_Schifffahrt/b_Verkehrsmanagement/2_Verkehrsgruppen/Verkehrsgruppen_node.html) [Accessed at 26.03.2025]
[2] Danish Historic AIS-Data (<https://www.dma.dk/safety-at-sea/navigational-information/ais-data>) [Accessed at 26.03.2025]
[3] AIS-Stream for live map (<https://aisstream.io/>) [Accessed at 26.03.2025]
[4] <https://github.com/ameliee-creator/DataScience>

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